

Notes: Fisman (AER, 2001)

Estimating the Value of Political Connections

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1 Big picture

- **Question.** How valuable are political connections for firms? The paper asks this in a concrete way: how much of a firm's market value in Indonesia depended on its ties to President Suharto?
- **Setting.** Indonesia in the mid-1990s is ideal for this question because political decision-making was highly centralized around Suharto, and business groups' political exposure could be summarized in a reasonably credible index.
- **Core challenge.** One cannot simply compare profits or average firm values across connected and unconnected firms. Connected firms may spend resources on rent-seeking, and unobserved ability may be correlated with the ability to form political ties.
- **Main identification idea.** The paper uses **event-study variation** from rumors about Suharto's health. If a firm's value depends on Suharto staying in power, then adverse news about his health should hurt that firm's stock price more.
- **Main empirical result.** In every major Suharto-health episode, politically dependent firms had lower stock returns than less-dependent firms. In the pooled regression, this differential effect becomes stronger when the market reaction to the rumor is more severe.
- **Key coefficient.** In the pooled specification, the interaction between political dependence and event severity is positive and statistically significant, with coefficient about **0.28**. Since severe events are associated with more negative net market returns, this means that more connected firms suffer *more* during worse rumors.
- **Back-of-the-envelope value.** Using bankers' view that a sudden Suharto death would have lowered the Jakarta index by roughly **20%**, the paper infers a coefficient on political dependence of about **5.8**. This implies that a firm with the highest political dependence would have returns about **23 percentage points** lower than an otherwise similar unconnected firm in such a regime-shift event.
- **Main economic interpretation.** A substantial part of well-connected firms' market value appears to come from expected political favors rather than purely from underlying productivity.
- **Why this paper matters.** It is one of the classic papers showing how political connections can be valued using financial markets. It also provides an early clean event-study design for estimating private political rents at the firm level.

2 Data and measurement (key objects)

2.1 Accounting and share-price data

- Accounting data come from the *Financial Times* Extel Financials Database.
- The paper uses 1995 accounting data because that year offers the best recent coverage.
- Key firm-level accounting variables are:
 - total assets, denoted **ASSETS**,
 - total debt, denoted **DEBT**,

- taxes paid, denoted **TAX**,
- net income, denoted **NI**,
- the firm’s industry classification.
- Stock-price data come mainly from the *Financial Times* Extel Securities Database, with holes filled using Investamatic data.
- The firms are publicly traded firms on the Jakarta Stock Exchange (JSX).

2.2 Group affiliation data

- Indonesian group affiliations are not publicly reported in a single official dataset, so they must be pieced together from ownership, board, and management information.
- The main source is *Top Companies and Big Groups in Indonesia* (Kompas Indonesia, 1996), which lists the top 200 Indonesian groups and their affiliates.
- For newer firms, the *Indonesian Capital Markets Directory 1997* is used.
- Firms affiliated with the Tahija group are identified from a recent *AsiaMoney* article.
- All classifications are checked with investment analysts in Jakarta in December 1997.
- A useful feature of the setting is that group affiliations among listed companies are essentially stable over the period, so the paper does not need to model time-varying ownership shifts.

2.3 Political connectedness: the Suharto Dependency Index

- The paper’s core political variable is the **Suharto Dependency Index**, constructed by the Castle Group, a Jakarta consulting firm.
- The index rates each of the 25 largest industrial groups by the degree to which it depends on political connections for its profitability.
- The raw score ranges from:

1 (least dependent) to 5 (most dependent).

- These 25 groups contain multiple listed companies, yielding a sample of **79 firms**.
- Groups owned by Suharto’s children and close allies receive the highest values; groups such as the Bakrie brothers’ and Julius Tahija’s are at the low end.
- In regressions, the paper subtracts one from the raw rating to simplify interpretation of interaction terms, so the regression version of POL_i runs from 0 to 4.

2.4 Health-rumor events

- The paper identifies episodes in which financial markets reacted to rumors about Suharto’s health.
- Events are found using a Lexis-Nexis search with the keywords **SUHARTO**, **HEALTH**, **INDONESIA**, and either **STOCK** or **FINANCIAL**.

- This returns 484 stories, most of which refer to one or more of six episodes.
- The event window begins with the specific triggering event that put Suharto’s health in question.
- The event ends when either:
 - new information explicitly resolves the rumor, or
 - the press reports that analysts have already priced the news into securities.
- The six event windows are:
 - January 30–February 1, 1995,
 - April 27, 1995,
 - April 29, 1996,
 - July 4–9, 1996,
 - July 26, 1996,
 - April 1–3, 1997.

2.5 Key measured objects

- **Firm return during event e :** the stock-price return of firm i over event window e , denoted R_{ie} .
- **Political dependence:** the firm’s transformed Suharto Dependency Index, denoted POL_i .
- **Event severity:** the return on the Jakarta Stock Exchange Composite Index net of broader Southeast Asian market movements, denoted $NR_e(JCI)$.
- **Pooled sample size:** 455 firm-event observations in the main pooled regression.

2.6 Summary statistics

The paper reports summary statistics by degree of political dependence. The main message is that firms with different dependence scores do *not* look systematically different in obvious balance-sheet dimensions.

- There are 79 affiliated firms in total.
- Available observations are 76 for assets and return on assets, 70 for debt, and 74 for the tax-rate calculation.
- Mean assets are large in all groups, and there is no clear monotonic relationship between political dependence and firm size.
- The same is true for debt, return on assets, and tax rates.

2.7 Unobservables and timing

- The core unobservable is the true private value of political access.

- The event-study design treats rumors about Suharto’s health as shocks to the expected continuation value of the political regime.
- Because the paper studies short event windows, it abstracts from slow-moving fundamentals and focuses on immediate valuation effects.

3 Models and empirical specifications (all equations + interpretation)

3.1 Event-specific cross-sectional regression

The first regression is run separately for each event:

$$R_{ie} = \alpha + \rho POL_i + \varepsilon_{ie}. \quad (1)$$

- **Dependent variable.** R_{ie} is firm i ’s stock return during event e .
- **Regressor.** POL_i measures how politically dependent the firm is.
- **Interpretation.** A negative ρ means that more connected firms lose more value when Suharto’s health is called into question.
- **Economic meaning.** This is the most direct test of whether politically connected firms are more exposed to bad news about the continuation of Suharto’s regime.

3.2 Net market return as a measure of event severity

Because each rumor differs in severity, the paper constructs an event-level measure of how seriously investors took the rumor. First, it estimates a **market model** for daily Jakarta returns in 1994:

$$R_t(JCI) = \alpha + \sum_{m \in M} \beta_m R_t(m) + \varepsilon_t, \quad (2)$$

where:

- $R_t(JCI)$ is the daily return on the Jakarta Composite Index,
- $R_t(m)$ is the return on regional market index m ,
- M includes Tokyo’s Nikkei 225, Hong Kong’s Hang Seng, Singapore’s Straits Times, Bangkok’s SET, Taiwan’s Weighted, the Philippines’ Composite, Kuala Lumpur’s Composite, and Seoul’s Composite.

Using the estimated coefficients from (2), the paper defines the event-level net return of the Jakarta index as:

$$NR_e(JCI) = R_e(JCI) - \left[\hat{\alpha} + \sum_{m \in M} \hat{\beta}_m R_e(m) \right]. \quad (3)$$

- **Interpretation.** $NR_e(JCI)$ removes broader Southeast Asian market movements and isolates the part of the Jakarta market decline specific to the Suharto-health episode.

- **Event severity logic.** More negative values of $NR_e(JCI)$ correspond to more serious rumors, because investors interpreted them as more threatening to regime continuation.

3.3 Pooled interaction regression

The paper then pools all events and allows the connectedness effect to vary with event severity:

$$R(P_{ie}) = \alpha + \rho_1 POL_i + \rho_2 NR_e(JCI) + \rho_3 [NR_e(JCI) \cdot POL_i] + \varepsilon_{ie}. \quad (4)$$

- **Why pool?** Pooling uses all six events at once and tests whether the connectedness effect is systematically stronger in worse rumors.
- **Marginal effect of political dependence.** From (4),

$$\frac{\partial R(P_{ie})}{\partial POL_i} = \rho_1 + \rho_3 NR_e(JCI). \quad (5)$$

- Since severe rumors correspond to more negative $NR_e(JCI)$, a positive ρ_3 implies that the marginal effect of political dependence becomes *more negative* as event severity increases.
- **Estimated coefficient.** The paper finds $\hat{\rho}_3 \approx 0.28$, statistically significant at the 5 percent level.
- **Interpretation.** If the overall market declines by 1 percent in response to Suharto-health news, then a firm with $POL = x$ is predicted to fall about 0.28 percentage points more than a firm with $POL = x - 1$.

3.4 Event-study scaling and the value of connections

The actual rumors only increased the probability that Suharto might leave office; they were not full regime-shift events. The paper therefore uses equation (4) to extrapolate what the effect would be if Suharto died suddenly.

During a visit to Jakarta in August 1997, Fisman asked investment bankers how much they thought the Jakarta Composite Index would fall if Suharto died unexpectedly. The modal answer was **20 percent**. Plugging this into equation (4), the paper infers that the coefficient on political dependence would be about:

$$\hat{\rho}_1 + \hat{\rho}_3(-20) \approx -0.19 + 0.28(-20) \approx -5.8. \quad (6)$$

Hence the difference between a maximally dependent firm ($POL = 4$) and an unconnected firm ($POL = 0$) would be approximately:

$$4 \times 5.8 \approx 23 \text{ percentage points}. \quad (7)$$

- **Interpretation.** This is the paper's back-of-the-envelope estimate of how much market value highly connected firms stood to lose in a genuine regime break.
- **Caution.** The paper explicitly notes that this inference is far out of sample, because no truly sudden and clean regime shift occurred during the period studied.

3.5 Flexible monotonicity specification

To check whether a linear treatment of POL is reasonable, the paper estimates a more flexible specification:

$$R_{ie} = \alpha + \sum_{p=2}^5 \beta_p I_{POL \geq p} + \sum_{e=2}^6 \gamma_e + \varepsilon_{ie}, \quad (8)$$

where

$$I_{POL \geq p} = \begin{cases} 0 & \text{if } POL < p, \\ 1 & \text{if } POL \geq p. \end{cases} \quad (9)$$

- **Interpretation.** Instead of forcing a single linear slope, this allows returns to vary flexibly with the level of political dependence.
- **Result.** The coefficients are uniformly negative, ranging from roughly -0.35 to -0.87 , and the equality of the β_p 's cannot be rejected.
- **Implication.** The linear specification for POL is a reasonable summary of the data.

3.6 Standard errors and controls

- The paper reports heteroskedasticity-robust standard errors.
- It also checks specifications allowing correlation of errors within firms and within business groups; the conclusions are very similar.
- Additional controls such as $\log(ASSETS)$, $\log(DEBT)$, and industry dummies do not materially alter the interaction result.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive approach

- Comparing average profitability across connected and unconnected firms does not identify the value of political ties.
- In equilibrium, connected firms may not look more profitable if they spend heavily on rent-seeking.
- More fundamentally, unobservables such as business skill may be correlated with connectedness.

4.2 What identifies the paper's strategy

- The event-study approach differences out many slow-moving firm characteristics and focuses on **changes in value** around short political shocks.
- The source of identification is cross-sectional heterogeneity in exposure to a common political shock.
- If adverse news about Suharto hurts connected firms more than less-connected firms, that differential market reaction can be interpreted as evidence that connected firms derive value from the regime.

4.3 Why Indonesia is especially useful

- Political power was unusually centralized and stable, so the relevant political connection was concentrated in one person and his network.
- This makes it possible to construct a meaningful summary index of political dependence.
- In a more decentralized political system, this sort of clean measurement would be much harder.

4.4 Why the severity measure matters

- One might worry that individual events differ idiosyncratically and that the connectedness effect is just noise.
- The net Jakarta return, $NR_e(JCI)$, provides an observable event-level statistic summarizing how threatening the rumor was to the regime.
- The almost perfect correlation between the event-specific *POL* coefficient and $NR_e(JCI)$ is therefore important: it shows that the effect scales with event seriousness in exactly the predicted way.

4.5 Key threats and how the paper addresses them

- **Thin trading.** If a stock does not trade, measured returns can be zero even when the underlying value falls. Restricting the sample to firms with positive trading volume during the event changes the main interaction result only slightly.
- **Higher bad-news beta for connected firms.** Maybe connected firms simply perform worse on any bad news day. The paper checks this by looking at the start of the Southeast Asian crisis and large negative S&P 500 days. It finds no relationship between returns and political dependence in those episodes.
- **Event-window misspecification.** Trading on inside information could move prices before the official rumor date. Expanding the event window backward by one or two days only mildly attenuates the results.
- **Nonlinearity in the political-dependence index.** The flexible indicator specification confirms a monotone negative relationship.

4.6 What the paper cannot identify perfectly

- The observed rumors are not equivalent to a clean, unexpected regime collapse.
- Suharto's actual removal in May 1998 cannot be used cleanly because it coincided with currency collapse, riots, political instability, an IMF program, thin trading, and uncertainty about whether Habibie represented a true regime change.
- Therefore, the paper identifies the value of political ties through rumor-induced changes in continuation probabilities, not through a full regime-switch experiment.

4.7 Relation to the literature

- Relative to **Krueger (1974)**, this paper moves from aggregate rent-seeking to the valuation of rents at the firm level.
- Relative to **Roberts (1990)**, it studies an environment where political ties plausibly account for a much larger fraction of firm value.
- More broadly, it is an early finance-meets-political-economy paper using security prices to value political access.

5 Tables and figures

5.1 Table 1: Summary statistics by degree of political dependence

Read:

- The sample contains **79 firms** affiliated with politically relevant Indonesian business groups.
- There are no obvious systematic differences across connectedness categories in firm size, leverage, return on assets, or tax rates.
- This matters because the later event-study results are less likely to be driven by connected firms simply being mechanically different in balance-sheet terms.
- The paper is therefore not just picking up the fact that "big firms" or "high-debt firms" react differently.

TABLE 1—SUMMARY STATISTICS BY DEGREE OF POLITICAL DEPENDENCE AS MEASURED BY THE SUHARTO DEPENDENCY INDEX

POL	1	2	3	4	5	All firms	Observations
Observations	5	34	10	16	14	79	
Assets	2,145.76 (2,843.63)	2,228.37 (3,989.85)	2,206.20 (3,676.99)	1,634.08 (2,561.07)	1,765.51 (2,230.52)	2,033.19 (3,321.59)	76
Debt	707.18 (702.84)	791.32 (1,478.83)	813.25 (976.28)	397.83 (461.06)	712.57 (1,070.83)	717.37 (1,186.85)	70
Return on assets (net income)/ (total assets)	0.038 (0.031)	0.058 (0.058)	0.043 (0.023)	0.037 (0.032)	0.050 (0.029)	0.050 (0.044)	76
Tax rate (taxes paid)/(pretax income)	0.23 (0.05)	0.24 (0.12)	0.16 (0.14)	0.22 (0.16)	0.15 (0.12)	0.21 (0.13)	74

Sources: All data are from the *Financial Times' Extel Database* (1997); Assets and Debt are expressed in millions of 1995 rupiah.

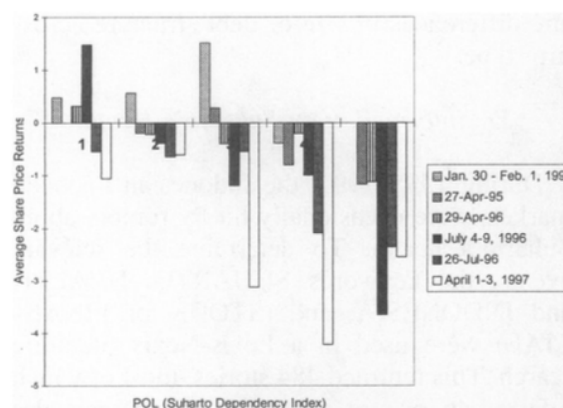


FIGURE 1. EFFECT OF POLITICAL DEPENDENCE ON SHARE PRICE RETURNS

5.2 Figure 1: Effect of political dependence on share-price returns

Read:

- For each of the six Suharto-health episodes, average stock returns are plotted against the Suharto Dependency Index.

- In every episode, the relationship slopes downward: more politically dependent firms perform worse.
- The slope is much steeper in the most serious events, especially July 4–9, 1996 and April 1–3, 1997.
- This picture is the raw visual motivation for the later pooled interaction regression.

5.3 Table 2: Event-specific regressions

Read:

- The coefficient on *POL* is negative in *every* event-specific regression.
- The point estimates are approximately:
 - −0.58 for Jan. 30–Feb. 1, 1995,
 - −0.31 for Apr. 27, 1995,
 - −0.24 for Apr. 29, 1996,
 - −0.95 for Jul. 4–9, 1996,
 - −0.57 for Jul. 26, 1996,
 - −0.90 for Apr. 1–3, 1997.
- The magnitudes vary across events, which is exactly what motivates modeling event severity.
- The correlation between these event-specific coefficients and the net Jakarta return is **0.98**, an extremely strong pattern.

TABLE 2—EFFECT OF POLITICAL CONNECTIONS ON CHANGES IN SHARE PRICE, SEPARATE ESTIMATION FOR EACH EVENT

	Jan. 30–Feb. 1, 1995	April 27, 1995	April 29, 1996	July 4–9, 1996	July 26, 1996	April 1–3, 1997
<i>POL</i>	−0.58* (0.34)	−0.31 (0.18)	−0.24* (0.15)	−0.95*** (0.27)	−0.57*** (0.22)	−0.90** (0.35)
Constant	1.29 (0.79)	0.21 (0.32)	0.12 (0.46)	0.83 (0.64)	−0.07 (0.41)	0.77 (0.97)
R^2	0.037	0.043	0.025	0.147	0.078	0.075
Observations	70	70	78	79	79	79

Note: Robust standard errors are in parentheses.

* Significantly different from 0 at the 10-percent level.

** Significantly different from 0 at the 5-percent level.

*** Significantly different from 0 at the 1-percent level.

TABLE 3—EFFECT OF POLITICAL CONNECTIONS ON CHANGES IN SHARE PRICE

	(1)	(2)
<i>POL</i>	−0.60** (0.11)	−0.19 (0.15)
<i>NR(JCI)</i>	0.25 (0.14)	−0.32 (0.28)
<i>NR(JCI) · POL</i>		0.28* (0.11)
Constant	0.88 (0.27)	0.06 (0.35)
R^2	0.066	0.078
Number of observations	455	455

Note: Robust standard errors are in parentheses.

* Significantly different from 0 at the 5-percent level.

** Significantly different from 0 at the 1-percent level.

5.4 Table 3: Pooled interaction regression

Read:

- Column (1) pools all events without the interaction and finds a negative average coefficient on *POL*, about −0.60.

- Column (2) adds the interaction term $NR(JCI) \times POL$ and finds:
 - $\hat{\rho}_1 \approx -0.19$,
 - $\hat{\rho}_2 \approx -0.32$,
 - $\hat{\rho}_3 \approx 0.28$.
- Because severe rumors imply more negative $NR(JCI)$, the positive interaction means politically connected firms lose more in worse episodes.
- This is the paper’s main statistical result.

5.5 Section III robustness checks

Read:

- **Thin trading:** restricting to stocks with positive trading volume during an event leaves the main result essentially unchanged.
- **General bad-news beta:** there is no relation between connectedness and returns during the 1997 rupiah float episode or after large S&P 500 drops.
- **Event-window choice:** adding one or two days prior to the event only slightly weakens the results.
- **Monotonicity:** the flexible specification produces uniformly negative coefficients from -0.35 to -0.87 , and equality of those coefficients cannot be rejected with $F(3, 445) = 0.24$.

6 Short summary

- **Conclusion.** The paper estimates the value of political connections by examining stock-market reactions to rumors about Suharto’s health.
- **Identification insight.** A firm’s stock price should respond more negatively to adverse political news the more dependent that firm is on the incumbent regime.
- **Main finding.** Politically dependent firms lost significantly more value during severe Suharto-health episodes, implying that political rents formed a substantial part of connected firms’ market value.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Political connections in Indonesia were not a sideshow. For a meaningful set of large listed firms, they were economically important enough to show up clearly in stock-market valuations.
- The event-study evidence indicates that the expected survival of Suharto’s regime materially affected the value of politically connected firms.
- The paper therefore provides direct evidence that political rents can be large at the firm level.

7.2 Economic intuition

- A connected firm is valuable not only because of its assets and productivity, but also because investors expect future favors: easier access to licenses, contracts, protection, or finance.
- Adverse health rumors reduce the probability that these favors will continue.
- The stock market capitalizes this probability change immediately, so connected firms decline more when investors fear the political regime may end.

7.3 Takeaway

This paper is a clean illustration of how financial markets can be used to value political capital. Its contribution is not just to show that political connections matter, but to show *how much* they matter in market value terms. The central message is that in environments with concentrated political power, a substantial share of firm value may rest on political access. For work on political economy, firm value, business groups, corruption, or crony capitalism, this paper is a benchmark event-study design.